

g C-2024

Status Finished

Solved Monday, 23 December 2024, 5:33 PM

Submitted Saturday, 23 November 2024, 11:08 AM

Time taken 30 days 6 hours

Write a program to read two integer values and print true if both the numbers end with the same digit, otherwise print false. Example: If 698 and 768 are given, program should print true as they both end with 8. Sample Input 1 25 53 Sample Output 1 false Sample Input 2 27 77 Sample Output 2 true

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int X,Y;
5     scanf("%d%d",&X,&Y);
6     if(X%10==Y%10)
7     {
8         printf("true");
9     }
10    else
11    {
12        printf("false");
13    }
14 }
```

Print Weird if the number is weird; otherwise, print Not Weird.

Sample Input 0

3

Sample Output 0

Weird

Sample Input 1

24

Sample Output 1

Not Weird

Explanation

Sample Case 0: $n = 3$

n is odd and odd numbers are weird, so we print *Weird*.

Sample Case 1: $n = 24$

$n > 20$ and n is even, so it isn't weird. Thus, we print *Not Weird*.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     if(n%2==0)
7     {
8         if(n>=3 && n<=5)
9         {
10            printf("Not Weird");
11        }
12    }
13    if(n>=6 && n<=20)
14    {
15        printf("Weird");
16    }
17    if (n>20)
18    {
19        printf("Not Weird");
20    }
21 }
22 else
23 {
24     printf("Weird");
25 }
26 }
27
28
```

Three numbers form a Pythagorean triple if the sum of squares of two numbers is equal to the square of the third. For example, 3, 5 and 4 form a Pythagorean triple, since $3^2 + 4^2 = 25 = 5^2$. You are given three integers, a, b, and c. They need not be given in increasing order. If they form a Pythagorean triple, then print "yes", otherwise, print "no". Please note that the output message is in small letters. Sample Input 1 3 5 4 Sample Output 1 yes Sample Input 2 5 8 2 Sample Output 2 no

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int a,b,c;
5      scanf("%d%d%d",&a,&b,&c);
6      if(a*a+b*b==c*c)
7      {
8          printf("yes");
9      }
10     else if(a*a+c*c==b*b)
11     {
12         printf("yes");
13     }
14     else if (b*b+c*c==a*a)
15     {
16         printf("yes");
17     }
18     else
19     {
20         printf("no");
21     }
22     return 0;
23 }
```

	Input	Expected	Got	
✓	3 5 4	yes	yes	✓
✓	5 8 2	no	no	✓

Passed all tests! ✓

```
1 #include <stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d", &n);
6     if (n==3)
7     {
8         printf("Triangle");
9     }
10    else if(n==4)
11    {
12        printf("Square");
13    }
14    else if(n==5)
15    {
16        printf("Pentagon");
17    }
18    else if(n==6)
19    {
20        printf("Hexagon");
21    }
22    else if(n==7)
23    {
24        printf("Heptagon");
25    }
26    else if(n==8)
27    {
28        printf("Octagon");
29    }
30    else if(n==9)
31    {
32        printf("Nonagon");
33    }
34    else if(n==10)
35    {
36        printf("Decagon");
37    }
38    else
39    {
40        printf("The number of sides is no
41    }
42 }
43
44
```

2000	Dragon
2001	Snake
2002	Horse
2003	Sheep
2004	Monkey
2005	Rooster
2006	Dog
2007	Pig
2008	Rat
2009	Ox
2010	Tiger
2011	Hare

Write a program that reads a year from the user and displays the animal associated with that year. Your program should work correctly for any year greater than or equal to zero, not just the ones listed in the table.

Sample Input 1

2004

Sample Output 1

Monkey

Sample Input 2

2010

Sample Output 2

Tiger

```

else if (year%12==3)
{
    printf("Sheep");
}
else if (year%12==0)
{
    printf("Monkey");
}
else if (year%12==1)
{
    printf("Rooster");
}
else if (year%12==2)
{
    printf("Dog");
}
else if (year%12==3)
{
    printf("Pig");
}
else if (year%12==4)
{
    printf("Rat");
}
else if (year%12==5)
{
    printf("Ox");
}
else if (year%12==6)
{
    printf("Tiger");
}
else
{

```

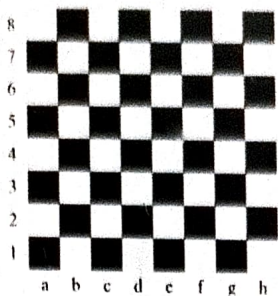

Question 3

Correct

Marked out of 2.00

Flag question

Positions on a chess board are identified by a letter and a number. The letter identifies the column, while the number identifies the row, as shown below:



Write a program that reads a position from the user. Use an if statement to determine if the column begins with a black square or a white square. Then use modular arithmetic to report the color of the square in that row. For example, if the user enters a1 then your program should report that the square is black. If the user enters d5 then your program should report that the square is white. Your program may assume that a valid position will always be entered. It does not need to perform any error checking.

Sample Input 1

a 1

Sample Output 1

The square is black.

Sample Input 2

d 5

Sample Output 2

The square is white.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int num,sum;
5     char alpha;
6     scanf("%c%d",&alpha,&num);
7     sum=alpha-num;
8     if(sum%2==0)
9     {
10         printf("The square is black.");
11     }
12     else
13     {
14         printf("The square is white.");
15     }
16 }
17
```

Input	Expected	Got
✓ a 1	The square is black.	The square is bl
✓ d 5	The square is white.	The square is wr
Passed all tests! ✓		

Some data sets specify dates using the year and day of year rather than the year, month, and day of month. The day of year (DOY) is the sequential day number starting with day 1 on January 1st.

There are two calendars - one for normal years with 365 days, and one for leap years with 366 days. Leap years are divisible by 4. Centuries, like 1900, are not leap years unless they are divisible by 400. So, 2000 was a leap year.

To find the day of year number for a standard date, scan down the Jan column to find the day of month, then scan across to the appropriate month column and read the day of year number. Reverse the process to find the standard date for a given day of year.

Write a program to print the Day of Year of a given date, month and year.

Sample Input 1

18
6
2020

Sample Output 1

170

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int d,m,y,feb;
5     scanf("%d%d%d",&d,&m,&y);
6     if((y%100!=0 && y%400==0)|| (y%4==0))
7         feb=29;
8     else
9         feb=28;
10    switch(m)
11    {
12        case 1:
13            printf("%d",d);
14            break;
15        case 2:
16            printf("%d",31+d);
17            break;
18        case 3:
19            printf("%d",31+feb+d);
20            break;
21        case 4:
22            printf("%d",31+feb+31+d);
23            break;
24        case 5:
25            printf("%d",31+feb+31+30+d);
26            break;
27        case 6:
28            printf("%d",31+feb+31+30+31+d);
29            break;
30        case 7:
31            printf("%d",31+feb+31+30+31+30+d);
32            break;
33        case 8:
34            printf("%d",31+feb+31+30+31+30+31);
35            break;
36        case 9:
37            printf("%d",31+feb+31+30+31+30+31);
38            break;
39        case 10:
40            printf("%d",31+feb+31+30+31+30+31);
41            break;
42        case 11:
43            printf("%d",31+feb+31+30+31+30+31);
44            break;
45        case 12:
46            printf("%d",31+feb+31+30+31+30+31);
47            break;
48    }
```



```
0.
1. %d, %m, %y),
2. %d, %m, %y),
3. %d, %m, %y),
4. %d, %m, %y),
5. %d, %m, %y),
6. %d, %m, %y),
7. %d, %m, %y),
8. %d, %m, %y),
9. %d, %m, %y),
10. %d, %m, %y),
11. %d, %m, %y),
12. %d, %m, %y),
13. %d, %m, %y),
14. %d, %m, %y),
15. %d, %m, %y),
16. %d, %m, %y),
17. %d, %m, %y),
18. %d, %m, %y),
19. %d, %m, %y),
20. %d, %m, %y),
21. %d, %m, %y),
22. %d, %m, %y),
23. %d, %m, %y),
24. %d, %m, %y),
25. %d, %m, %y),
26. %d, %m, %y),
27. %d, %m, %y),
28. %d, %m, %y),
29. %d, %m, %y),
30. %d, %m, %y),
31. %d, %m, %y),
32. %d, %m, %y),
33. %d, %m, %y),
34. %d, %m, %y),
35. %d, %m, %y),
36. %d, %m, %y),
37. %d, %m, %y),
38. %d, %m, %y),
39. %d, %m, %y),
40. %d, %m, %y),
41. %d, %m, %y),
42. %d, %m, %y),
43. %d, %m, %y),
44. %d, %m, %y),
45. %d, %m, %y),
46. %d, %m, %y),
47. %d, %m, %y),
48. %d, %m, %y),
49. %d, %m, %y),
```

```
0. %d", d);
```

```
1. %d", 31+d);
```

```
2. %d", 31+feb+d);
```

```
3. %d", 31+feb+31+d);
```

```
4. %d", 31+feb+31+30+d);
```

```
5. %d", 31+feb+31+30+31+d);
```

```
6. %d", 31+feb+31+30+31+30+d);
```

```
7. %d", 31+feb+31+30+31+30+31+d);
```

```
8. %d", 31+feb+31+30+31+30+31+31+d);
```

```
9. %d", 31+feb+31+30+31+30+31+31+30+d);
```

```
10. %d", 31+feb+31+30+31+30+31+31+30+31+d);
```

```
11. %d", 31+feb+31+30+31+30+31+31+30+31+30+d);
```

Sample Output 4

0

Explanation

First is output of area of rectangle

Then, output of area of triangle

Then output of area square

Finally, something random, so we print 0

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b;
5     char c;
6     scanf("%c%d%d",&c,&a,&b);
7     switch(c)
8     {
9         case 'R':
10            printf("%d",a*b);
11            break;
12         case 'S':
13            printf("%.0f",(0.5)*a*b);
14            break;
15         case 'T':
16            printf("%d",a*b);
17            break;
18         default:
19            printf("0");
20            break;
21     }
22 }
```

```
1 #include<stdio.h>
2 int main()
3 {
4     int n ,day;
5     scanf("%d",&n);
6     if(n<296)
7         day=n;
8     else
9         day=n-296;
10    day%=10;
11    day=day+1;
12    day%=10;
13    switch(day)
14    {
15        case 1:
16            printf("Sunday");
17            break;
18        case 2:
19            printf("Monday");
20            break;
21        case 3:
22            printf("Tuesday");
23            break;
24        case 4:
25            printf("Wednesday");
26            break;
27        case 5:
28            printf("Thursday");
29            break;
30        case 6:
31            printf("Friday");
32            break;
33        case 7:
34            printf("saturday");
35            break;
36        case 8:
37            printf("Kryptonday");
38            break;
39        case 9:
40            printf("Coluday");
41            break;
42        case 10:
43            printf("Daxamday");
44            break;
45    }
46 }
47 }
```